

The empirical recurrence for OEIS A197051, proved by transfer matrix

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Abstract

OEIS A197051 counts arrays of width 5 in which every entry is equal to the NUMBER of its own neighbours, in named directions, that stand in a stated relation to it. The entry carries an empirical recurrence of order 13 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is self-referential but local: a cell's neighbours lie in the row above, its own row and the row below, so a window of three consecutive rows decides the condition for the middle one, and the count is a walk count on $S = 71947$ states.

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1 The conjecture

OEIS A197051 is “Number of $n \times 5$ 0..4 arrays with each element equal to the number its horizontal and vertical zero neighbors.”. It has offset 1 and begins

$$4, 10, 38, 108, 358, 1132, 3580, 11382, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = a(n-1) + 4*a(n-2) + 10*a(n-3) + 4*a(n-4) - 20*a(n-5) + a(n-6) - 2*a(n-7) + 2*a(n-8) - 16*a(n-9) + 4*a(n-10) + a(n-13) \text{ for } n > 14.$$

As of the “Last modified” line on the live entry (Oct 24 2024 03:13:53, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 5 cells across. The NEIGHBOURS of a cell are the cells reached from it by the offsets

$$(-1, 0), (0, -1), (0, 1), (1, 0),$$

those that lie inside the array. The condition is that every entry counts its own neighbours that are zero: writing $v = x(i, j)$,

$$v = \#\{\text{neighbours } y \text{ of } (i, j) : y = 0\}.$$

3 Three rows decide one

Lemma 1. *Every offset above changes the row index by -1 , 0 or $+1$, so all the neighbours of a cell of row i lie in rows $i-1$, i and $i+1$. The condition for the whole of row i is therefore decided by those three rows, an absent row contributing no neighbours.*

Proof. Immediate from the offsets, together with the rule that a neighbour outside the array is not a neighbour. $\square$

So the state is the pair (row above, current row), the row above allowed to be the symbol $\perp$; a step appends a row and settles the condition for the middle row of the window it completes. Two boundaries have to be right and neither is assumed. The walk starts at the state $(\perp, \perp)$, which stands for the empty array, so a walk of no steps is the empty array and one step is a one-row array. And the condition for the LAST row is settled by the end vector, which asks whether that row satisfies its condition with nothing written below it: the end vector is therefore NOT the all-ones vector, and a row that counts correctly only when something is written below it does not finish an array. With M the adjacency matrix,

$$a(n) = \iota^\top M^j \tau, \quad S = 71947.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^{13} - \sum_i c_i t^{13-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+13} - \sum_i c_i M^{j+13-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = a(n-1) + 4a(n-2) + 10a(n-3) + 4a(n-4) - 20a(n-5) + a(n-6) - 2a(n-7) + 2a(n-8) - 16a(n-9) + 4a(n-10)$$

for every $n > 14$.

Proof. The vector $w = q(M)\tau$ was formed by 13 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 14 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 25 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, count each cell's neighbours satisfying the relation, and keep the array when every cell equals its own count. No window, no transfer matrix and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.